

Let us first add some more data to create an address book. Let us add a section called addressbook in our LDAP tree under the base suffix. This section can be denoted as an Organisational Unit (OU). Therefore, our LDIF file will be:

```
[root@vbg ~]# cat /tmp/addressbook-base.ldif
dn: ou=addressbook,dc=knafl,dc=org
ou: addressbook
objectClass: top
objectClass: organizationalUnit
```

Let us add this OU to our directory:

```
[root@vbg ~]# ldapadd -x -D "cn=Manager,dc=knafl,dc=org" \
-W -f /tmp/addressbook-base.ldif
Enter LDAP Password:
adding new entry "ou=addressbook,dc=knafl,dc=org"
```

```
[root@vbg ~]#
```

Now let us add some address book entries under the OU 'addressbook'. An example of an LDIF file with a user entry might look like the file displayed below:

```
[root@vbg ~]# cat /tmp/addressbook.ldif
dn: cn=Varad Gupta,ou=addressbook,dc=knafl,dc=org
cn: Varad Gupta
sn: Gupta
l: Gurgaon
street: M-37, Old DLF Colony, Sector-14
st: Haryana
postalCode: 122003
homePhone: 0124 2222222
mobile: 0999999999
mail: varad.gupta@fosteringlinux.com
objectClass: top
objectClass: inetOrgPerson
[root@vbg ~]#
```

Add this entry to our directory:

```
[root@vbg ~]# ldapadd -x -D "cn=Manager,dc=knafl,dc=org" \
-w FOSTERINGLinux -f /tmp/addressbook.ldif
adding new entry "cn=Varad Gupta,ou=addressbook,dc=knafl,dc=org"
[root@vbg ~]#
```

Note the `-w` followed by the password of the rootdn instead of the `-W` used earlier. With the `-w` option, you can specify the password on the command line instead of being prompted for it.

You can similarly add other entries. Let us now query our directory from the Fedora client host:

```
[root@vbg ~]# ldapsearch -x -h 192.168.122.1 -b "dc=knafl,dc=org"
# extended LDIF
```

```
#
# LDAPv3
# base <dc=knafl,dc=org> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
```

```
# knafl.org
dn: dc=knafl,dc=org
dc: knafl
objectClass: top
objectClass: domain
```

```
# addressbook, knafl.org
dn: ou=addressbook,dc=knafl,dc=org
ou: addressbook
objectClass: top
objectClass: organizationalUnit
```

```
# Varad Gupta, addressbook, knafl.org
dn: cn=Varad Gupta,ou=addressbook,dc=knafl,dc=org
cn: Varad Gupta
sn: Gupta
l: Gurgaon
street: M-37, Old DLF Colony, Sector-14
st: Haryana
postalCode: 122003
homePhone: 0124 2222222
mobile: 0999999999
mail: varad.gupta@fosteringlinux.com
objectClass: top
objectClass: inetOrgPerson
#
# search result
search: 2
result: 0 Success
#
numResponses: 5
numEntries: 4
[root@vbg ~]#
```

In the next article, we will look at how to secure our basic LDAP server—adding an encrypted rootdn password, implementing access control, disallowing anonymous search, modifying and deleting entries, etc. If you have any queries, please feel free to mail me at varad.gupta@fosteringlinux.com. 

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